

## KROSNO, Poland

WMO: 126700

Lat: 49.7069N

Lon: 21.7692E

Elev: 1087

StdP: 14.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 4.8        | 10.4 | -0.4                          | 5.6 | 6.7  | 4.8 | 7.3 | 12.2 | 25.4                  | 34.7 | 22.3 | 33.3 | 5.0                   | 100  | 0.538 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.9                   | 84.2            | 68.5 | 81.1 | 66.9 | 78.3 | 65.5 | 70.1                | 80.8 | 68.4 | 78.4 | 66.9 | 75.9 | 7.3                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.4                            | 101.1 | 75.3 | 64.9 | 95.9 | 73.1 | 63.5 | 91.4 | 71.3 | 34.7          | 81.0 | 33.3 | 78.5 | 32.1 | 76.2 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 20.4          | 17.7 | 15.6 | DB                         | -1.3 | 88.3               | 8.4 | 2.4                                                | -7.4 | 90.0       | -12.3 | 91.4       | -17.1 | 92.8       | -23.2 | 94.5 | (4) |
| (5)               |      |      | WB                         | -2.4 | 72.7               | 8.0 | 1.4                                                | -8.1 | 73.7       | -12.7 | 74.5       | -17.2 | 75.3       | -23.0 | 76.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 47.9        | 27.4  | 30.0  | 37.8  | 49.0  | 56.7  | 62.8  | 66.1  | 65.8  | 56.9  | 49.0  | 40.9  | 31.0  | (6)  |  |
|                                                                        | DBStd       | 15.58       | 9.11  | 9.23  | 8.66  | 7.60  | 6.89  | 6.32  | 5.76  | 5.72  | 6.71  | 7.94  | 8.32  | 8.67  | (7)  |  |
|                                                                        | HDD50       | 2779        | 700   | 560   | 384   | 105   | 19    | 1     | 0     | 0     | 15    | 119   | 288   | 588   | (8)  |  |
|                                                                        | HDD65       | 6500        | 1165  | 980   | 844   | 482   | 268   | 115   | 59    | 59    | 255   | 497   | 723   | 1053  | (9)  |  |
|                                                                        | CDD50       | 2002        | 0     | 0     | 5     | 75    | 227   | 385   | 499   | 488   | 221   | 88    | 14    | 0     | (10) |  |
|                                                                        | CDD65       | 247         | 0     | 0     | 0     | 1     | 10    | 49    | 93    | 82    | 11    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 1840        | 0     | 0     | 0     | 13    | 105   | 363   | 682   | 613   | 63    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 379         | 0     | 0     | 0     | 0     | 6     | 66    | 150   | 151   | 6     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.3         | 8.4   | 8.3   | 8.5   | 7.7   | 6.9   | 6.4   | 6.1   | 5.8   | 6.1   | 7.1   | 8.2   | 8.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 33.3        | 2.0   | 1.8   | 1.9   | 2.3   | 3.9   | 4.0   | 5.1   | 3.3   | 3.0   | 2.3   | 2.0   | 1.7   | (15) |  |
|                                                                        | PrecMax     | 46.9        | 4.5   | 3.9   | 3.3   | 3.9   | 10.6  | 6.5   | 9.4   | 6.2   | 7.4   | 5.3   | 4.6   | 3.5   | (16) |  |
|                                                                        | PrecMin     | 24.3        | 0.8   | 0.6   | 0.8   | 0.9   | 1.9   | 1.3   | 1.5   | 0.9   | 0.9   | 0.2   | 0.1   | 0.6   | (17) |  |
|                                                                        | PrecStd     | 5.1         | 0.9   | 0.7   | 0.7   | 0.9   | 1.9   | 1.7   | 2.3   | 1.5   | 1.7   | 1.4   | 1.1   | 0.8   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 47.3  | 51.7  | 62.1  | 75.8  | 80.6  | 85.4  | 87.7  | 88.5  | 79.1  | 71.3  | 61.1  | 51.1  | (19) |  |
|                                                                        |             | MCWB        | 43.6  | 44.9  | 49.3  | 58.5  | 63.0  | 69.4  | 69.5  | 68.9  | 64.3  | 58.4  | 53.0  | 45.8  | (20) |  |
|                                                                        | 2%          | DB          | 44.0  | 47.1  | 57.3  | 70.3  | 77.1  | 81.7  | 84.0  | 84.0  | 75.7  | 67.4  | 57.7  | 48.3  | (21) |  |
|                                                                        |             | MCWB        | 41.2  | 41.7  | 47.2  | 55.4  | 62.2  | 67.7  | 68.9  | 68.0  | 63.5  | 58.3  | 51.0  | 44.3  | (22) |  |
|                                                                        | 5%          | DB          | 41.4  | 44.0  | 53.6  | 65.9  | 73.6  | 78.2  | 80.8  | 80.3  | 72.0  | 63.9  | 55.1  | 44.7  | (23) |  |
|                                                                        |             | MCWB        | 38.8  | 40.2  | 45.4  | 53.3  | 61.1  | 65.8  | 67.3  | 67.2  | 61.6  | 56.8  | 49.4  | 41.4  | (24) |  |
|                                                                        | 10%         | DB          | 38.8  | 41.3  | 50.1  | 62.0  | 69.7  | 74.9  | 77.8  | 76.9  | 68.5  | 60.6  | 52.4  | 41.5  | (25) |  |
|                                                                        |             | MCWB        | 36.7  | 38.5  | 43.6  | 51.4  | 59.0  | 64.0  | 65.9  | 65.5  | 59.8  | 54.9  | 48.0  | 39.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 44.1  | 45.7  | 51.0  | 59.7  | 66.6  | 71.3  | 72.7  | 71.9  | 66.4  | 61.3  | 54.3  | 46.7  | (27) |  |
|                                                                        |             | MCDB        | 46.2  | 50.9  | 58.3  | 72.6  | 77.1  | 82.5  | 83.8  | 83.4  | 76.3  | 67.3  | 58.8  | 50.2  | (28) |  |
|                                                                        | 2%          | WB          | 41.8  | 42.5  | 48.5  | 56.7  | 63.7  | 68.9  | 70.2  | 69.6  | 64.2  | 59.1  | 52.0  | 44.6  | (29) |  |
|                                                                        |             | MCDB        | 43.8  | 45.9  | 55.2  | 67.8  | 73.9  | 79.4  | 80.7  | 80.5  | 73.4  | 65.5  | 56.4  | 48.1  | (30) |  |
|                                                                        | 5%          | WB          | 39.2  | 40.7  | 46.4  | 54.5  | 61.8  | 66.8  | 68.7  | 68.0  | 62.4  | 57.1  | 50.2  | 42.0  | (31) |  |
|                                                                        |             | MCDB        | 41.0  | 43.3  | 52.6  | 63.8  | 71.3  | 75.9  | 78.2  | 78.3  | 70.2  | 63.3  | 54.2  | 44.4  | (32) |  |
|                                                                        | 10%         | WB          | 36.8  | 38.8  | 44.3  | 52.4  | 60.0  | 65.0  | 66.9  | 66.5  | 60.6  | 55.2  | 48.2  | 39.6  | (33) |  |
|                                                                        |             | MCDB        | 38.5  | 41.1  | 49.4  | 60.5  | 68.6  | 72.8  | 75.4  | 75.2  | 67.3  | 60.3  | 52.0  | 41.3  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.4   | 8.8   | 12.5  | 16.4  | 16.8  | 16.4  | 16.9  | 17.1  | 14.6  | 12.2  | 8.6   | 6.9   | (35) |  |
|                                                                        |             | MCDBR       | 7.8   | 11.9  | 18.6  | 22.8  | 22.9  | 21.4  | 21.9  | 21.9  | 20.3  | 16.4  | 11.5  | 8.7   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 6.1   | 8.2   | 10.9  | 11.3  | 11.1  | 10.1  | 9.0   | 9.2   | 9.9   | 9.1   | 7.0   | 6.3   | (37) |  |
|                                                                        |             | MCWBR       | 7.6   | 10.8  | 16.8  | 20.9  | 20.9  | 19.5  | 19.5  | 19.9  | 17.9  | 15.5  | 10.4  | 8.3   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.310 | 0.335 | 0.378 | 0.432 | 0.421 | 0.424 | 0.443 | 0.437 | 0.402 | 0.368 | 0.343 | 0.307 | (40) |  |
|                                                                        |             | taud        | 2.266 | 2.229 | 2.218 | 2.130 | 2.214 | 2.239 | 2.199 | 2.217 | 2.274 | 2.348 | 2.344 | 2.323 | (41) |  |
|                                                                        | Ebn at Noon |             | 228   | 251   | 260   | 257   | 266   | 266   | 258   | 253   | 249   | 237   | 214   | 213   | (42) |  |
|                                                                        |             | Edh at Noon | 23    | 30    | 37    | 45    | 43    | 42    | 43    | 41    | 35    | 27    | 22    | 19    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 281   | 520   | 879   | 1280  | 1561  | 1664  | 1608  | 1481  | 1039  | 635   | 317   | 213   | (44) |  |
|                                                                        | RadStd      |             | 42    | 79    | 111   | 145   | 197   | 144   | 181   | 122   | 151   | 93    | 39    | 27    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (30 neighbors) |  | +1.01 | N/A     | N/A    | +1.76   | +0.90 | +0.79 | -227        | -315  | +148  | +61   |

Nomenclature: See separate page